

# Mobile Ad Hoc Networking: Location Based Multicast Research and Proposed Improvement

By: [REDACTED]  
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(Senior Project)

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## 1. Introduction

A mobile ad hoc network consists of wireless mobile hosts that communicate with each other in the absence of a fixed infrastructure. A few characteristics of the mobile ad hoc network (MANET) consists of having wireless connections, being self-organized, and having no fixed infrastructure. There are many people who can benefit from MANET technology, such as the military, law enforcement, as well as all automobile drivers. There are many approaches that attempt to make ad hoc networking efficient and optimal as possible. One algorithm that has been proposed is Multicast flooding. Flooding is said by many to be the simplest multicast routing algorithm. Flooding can be used to deliver packets to nodes within a location-based multicast group. [2] Another algorithm is the location-based multicast algorithm, which uses multicast flooding with a defined forwarding zone. There are several proposed schemes that attempt to improve accuracy of multicast delivery and the overhead of the multicast delivery, although there is a trade-off between accuracy and overhead. I also proposed a scheme that attempts to improve upon overhead, while still considering the accuracy of the multicast delivery.

### 1.1 Applications of Mobile Ad Hoc Networks

#### • Battlefield

One application can be seen in the battlefield. If a wireless base station is destroyed by an enemy communication could be lost. Since ad hoc networks are self-organizing, soldiers can still communicate by forwarding data from one device to another.

#### • Traveling

Ad hoc networking can also be used to deal with traveling hassles, such as picking up a boarding pass. With the ad hoc network, your wireless device will have the ability to communicate with the airline system, allowing the airlines to confirm a seat number that can then be sent to you via MANET.

#### • Around the house

There are also many household applications that can benefit from the use of ad hoc networking. A person's ad hoc device could communicate with their homes wireless devices to unlock doors, adjust temperature settings, or even turn off an alarm. The device could also be used by children, which the children can be given a different level of control.

#### • Cars

With mobile ad hoc networking car-to-car communication is possible. If for example there happens to be a car accident on the road in which you are traveling, an alert can be sent to your car via other cars in the ad hoc network. [4]

## 2. Prior Arts

There are many approaches that have been taken in order to improve optimization and efficiency. Several of the approaches include geocasting and multicasting. Geocasting is when a packet is sent to all nodes inside a specified region. Multicasting is sending packets from source to many nodes. A broader term for multicasting could be broadcasting, in which packets are sent from a source to all nodes. Another form of casting is unicasting, which is when a packet is sent from a source to one node.

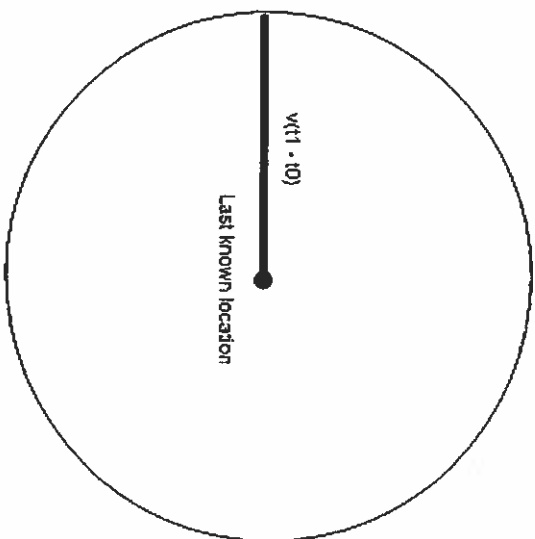
### 2.1 Unicasting

#### Location-Aided Routing (LAR)

LAR is a modified flooding algorithm that uses location information to improve performance of routing protocols for MANET, by using a search space for locating a route to the node. A Global Positioning System (GPS) may be used in order to provide location information to the LAR protocol. By implementing GPS, it is possible for a mobile host to know its physical location. It is assumed that each host knows its physical location precisely. Knowing the precise location allows us to have no error. LAR allows only nodes within the request zone to forward route requests, in which a request zone is explicitly specified in the route request. In order for a node to determine if it is in the request zone, it must know its physical location (with no error). Several advantages of using LAR is that it reduces overhead of route discovery as well as reduces the scope of route request flood. There are also several disadvantages. One disadvantage is that nodes need to know their physical location, and another is that it doesn't take into account the possible existence of obstructions for radio transmissions. LAR identifies several terms such as a request zone, and an expected zone that are explained below. [1]

- **Expected Zone**

An expected zone is a region that is expected to hold the current location of the destination. It is created by the node's last known geographical location, as well as the average speed of the node. Consider a node D to be a destination node, and a node S to be a source node. This method assumes that the source node S knows the location and the current time ( $t_0$ ) and average speed ( $v$ ) of the destination node, as seen in Figure 1. If the estimation is smaller than the actual average, then the destination could possibly lie outside of the expected zone at time  $t_1$ . This is why it is said that the expected zone only determines a region that might possess the destination at time  $t_1$ . If the previous location of node D is unknown, then the expected zone may be the size of the ad hoc network. In this case it is similar to the flooding algorithm. If however node S knows that node D is moving south, then the circular expected zone can be reduced into a smaller expected zone, such as a semi-circle. [1]



**Figure 1: Expected Zone Example**

- **Request Zone**

The request zone contains the expected zone and the location of the sender node. Only nodes that are located within the request zone can forward route requests. It is also explicitly specified in the route request. It is understood that each node must know its physical location in order to determine if it is in the request zone. If a route is not discovered in a relatively smaller request zone, a timeout will trigger the sender to initiate another route discovery using a larger request zone. Refer to figure 2 for an example of a request zone.

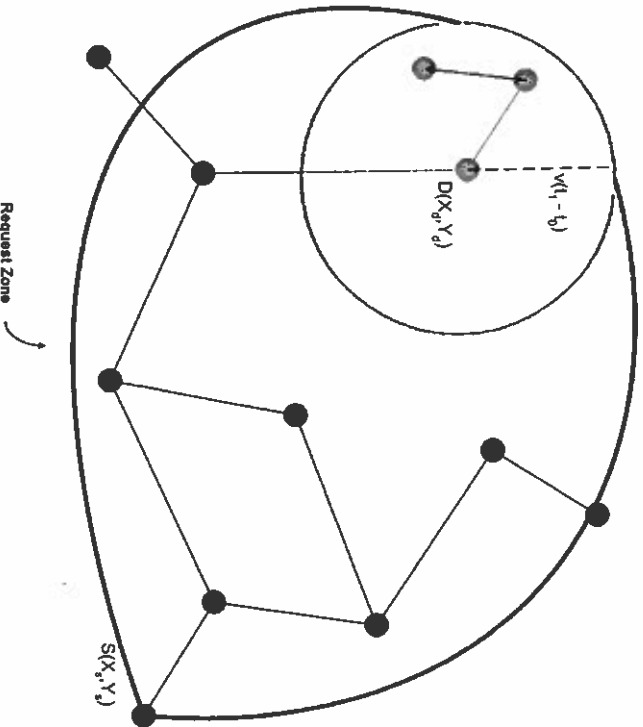


Figure 2: Request Zone Example

#### • Membership of Request Zones LAR Scheme 1

The first scheme proposed uses a rectangular shaped request zone. Scheme 1 assumes that the source node S knows the location of the destination node  $(X_d, Y_d)$  and the time of its last known location  $(t_0)$ . It is assumed that node S knows the average speed  $(v)$  of node D. With this, a circle of radius  $v(t_1 - t_0)$  and expected zone at time  $t_1$  is defined by node S. By using the above information, the source node can determine the four corners of the expected zone, thus allowing the source node to define the smallest rectangle that covers both the source node and the expected zone. When a route discovery message is transmitted by the source node, the coordinates of the rectangle are included within the message. During transmission, if an intermediate node I receives the route request from another node, node I forwards the request to its neighbors, because the intermediate node I

is located within the request zone. However, if a node J receives the route request and it determines that it is not located within the request zone, then node J simply discards the message without transmission. When the destination node D finally receives the route request message, it replies by sending a route reply message. The route reply message sent by D contains its current location, and the current time. When source node S receives the route reply message, it records the location of the destination node D. The information recorded by node S can be used to determine the request zone for a route discovery in the future. The size of the rectangular request zone mentioned above is proportional to the average speed of movement, and the time elapsed since the last known location of the destination was recorded, as seen in Figure 3. [1]

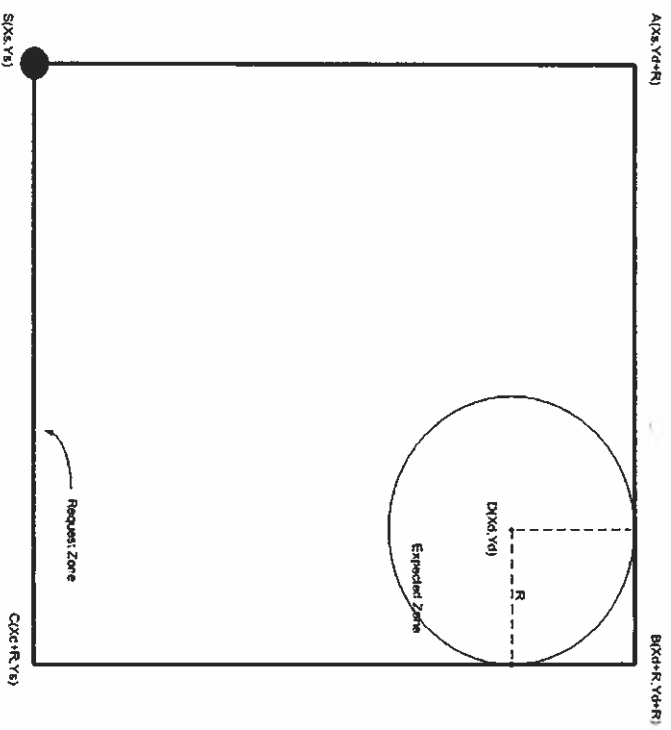


Figure 3: Source node outside of the expected zone

#### LAR Scheme 2

Recall that in scheme 1, the source node  $S$  created a request zone. In scheme 2, node  $S$  does not create a request zone, but rather computes its distance from the destinations location  $(X_d, Y_d)$ , and includes  $DIST_s$  in the route request message, along with the coordinates  $(X_s, Y_s)$ .  $DIST_s$  is the distance from node  $S$  to the destination node. When an intermediate node  $I$  receives the route request from the sender node, it calculates the distance from its location  $(X_s, Y_s)$  to the destination's location  $(X_d, Y_d)$ . The distance between  $(X_s, Y_s)$  and  $(X_d, Y_d)$  is denoted as  $DIST_I$ . If  $DIST_s + \delta \geq DIST_I$ , given that  $\delta$  is a defined parameter, then node  $I$  forwards the request to its neighbors, otherwise the message is discarded. When node  $I$  forwards the route request, it replaces the value of  $DIST_s$  with the new value  $DIST_I$ , along with the location of the destination  $(X_d, Y_d)$ . If an arbitrary node  $J$  receives the route request that it has already seen and processed, then node  $J$  simply discards the request. However, if it has not already seen and processed the request, it calculates its distance from  $(X_s, Y_s)$  which is denoted as  $DIST_J$ . If  $DIST_I + \delta \geq DIST_J$ , then node  $J$  forwards the request, otherwise it discards the message. [1]

## 2.2 Geocasting

### Location-Based Multicast (LBM)

Geocasting is a form of multicasting, in which the members are specified by a geographical region. The specified geographical region is referred to as the multicast region. The location-based multicast group is described as a set of nodes that reside within the specified multicast region. If a mobile node resides within a specified geographical region, then that node becomes a member of the corresponding multicast group. All of the geocasting schemes assume that every mobile node is equipped with a positioning system that knows its current location. [3]

#### • Multicast Flooding

Flooding is the simplest multicast routing algorithm used to deliver packets to nodes within a location-based multicast group. To implement a multicast flooding algorithm we need to assume that node  $S$  needs to send a packet to a specific multicast region, which is the circle in Figure 4. Node  $S$  (sender) then broadcasts the multicast packet to all of its surrounding neighbors. The various are all members of the multicast group members. If a node is within the specified multicast region, it will accept the packet, as well as broadcast the packet to its neighbors if it has not yet received the packet. If a node is located outside the multicast region and the packet has not been previously received, then the node broadcasts the packet to its neighbors, in hopes of finding members of the multicast region. [2]

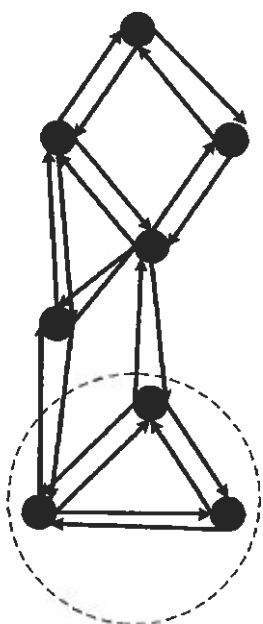


Figure 4: Multicast Flooding

#### • Multicast Region (with flooding)

Consider that the source node  $S$  needs to send a multicast packet to all nodes within a multicast region, which is typically represented by a closed-shaped polygon (see figure 5). Lets assume that nodes  $X$  and  $Y$  are in the multicast region at the time in which node  $S$  broadcasts a data packet, then the multicast group would contain nodes  $X$  and  $Y$ . The accuracy of delivery is the ratio of the number of nodes within the multicast region receiving the packet to the number of group members in the multicast region. For example, if there are 6 nodes located within the multicast region, and only 3 nodes received the multicast packet, then the accuracy of delivery would be 50%.

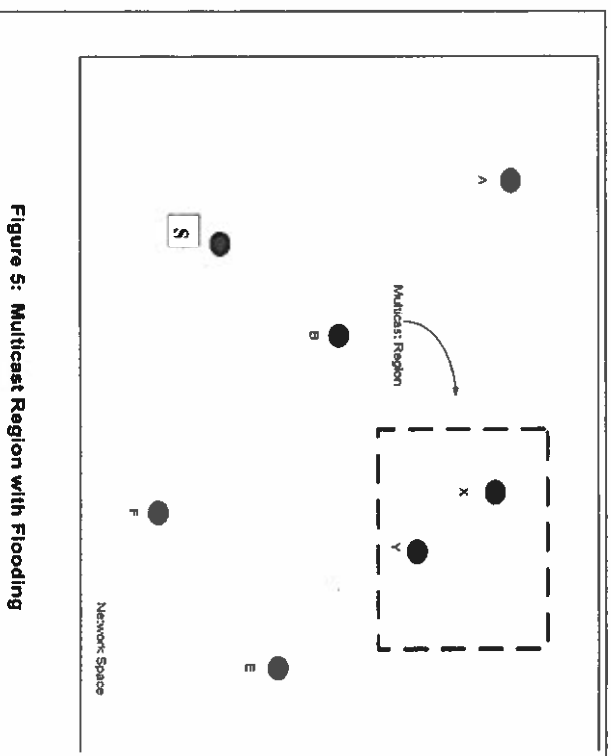


Figure 5: Multicast Region with Flooding

### • Forwarding Zone

Consider once again that a source node S needs to send a multicast packet to all nodes within a multicast region. One modification made to the LBM algorithm using flooding is that source node defines a forwarding zone for the multicast packet. What makes the forwarding zone different is that a node only forwards the multicast packet only if it is part of the forwarding zone. If the forwarding zone includes the multicast region, then the probability that a data packet will reach all its members increase. Although increasing the size of the forwarding zone increases accuracy, it has a downfall. By increasing the size of the forwarding zone, it causes data delivery overhead to increase.

### LBM Scheme 1

The first scheme that was proposed uses a rectangular shaped forwarding zone. Assume that the forwarding zone is the smallest rectangle that includes the current location of the source node S, as well as the multicast region. The sender node includes the coordinates of the corners of the rectangle in the multicast packet transmitted when it initiates a multicast delivery. If an intermediate node I

receives the multicast packet, and the node is not located within the specified rectangle, then the packet is discarded. Consider figure 6; if node I is located within the specified rectangle, it forwards the packet to its surrounding neighbors. It should be noted that the size of the rectangular forwarding zone is dependent on the size of the multicast region, as well as the location of the sender. A defined parameter  $\delta$  can be used to provide control of the forwarding zone size. The parameter  $\delta$  can also be used to extend the forwarding zone. When  $\delta$  is set to 0, the forwarding zone is identical to the multicast region, which can be seen in figure 7. [2]

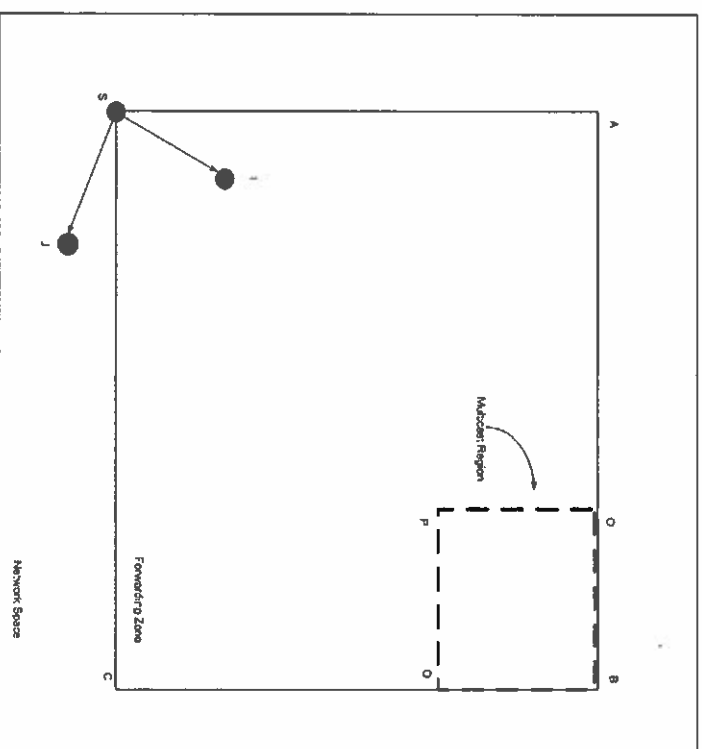


Figure 6: Source node outside of the multicast region (LBM Scheme 1)

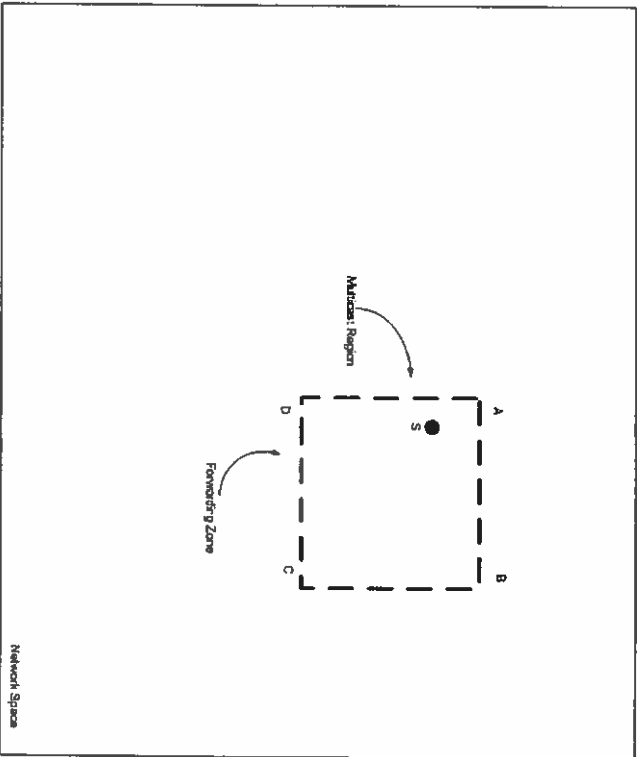


Figure 7: Source node inside the multicast region (LBM Scheme 1)

### LBM Scheme 2

The second scheme does not define the previously discussed forwarding zone, but rather the packet forwarding is determined by comparisons of the distance between the source node and center of the multicast region, as well as the distance between the packet receiving node and the center of the multicast region. The multicast packet contains three important pieces of information. First is the multicast region specification. Second is the location of the center of the multicast region  $(X_m, Y_m)$ , which the distance from a node  $I$  to  $(X_m, Y_m)$  is denoted as  $DIST_I$ . Finally the coordinates of the sender node  $(X_s, Y_s)$  will be included. When an intermediate node  $I$  receives the multicast packet, it determines if it belongs to the multicast region. If node  $I$  does belong to the region, then the distance from the source  $(X_s, Y_s)$  to node  $I$   $(X_i, Y_i)$  is calculated and denoted as  $DIST_I$ . If  $DIST_I + \delta \geq DIST_m$ , then node  $I$  forwards the packet to its neighbors, else it discards the

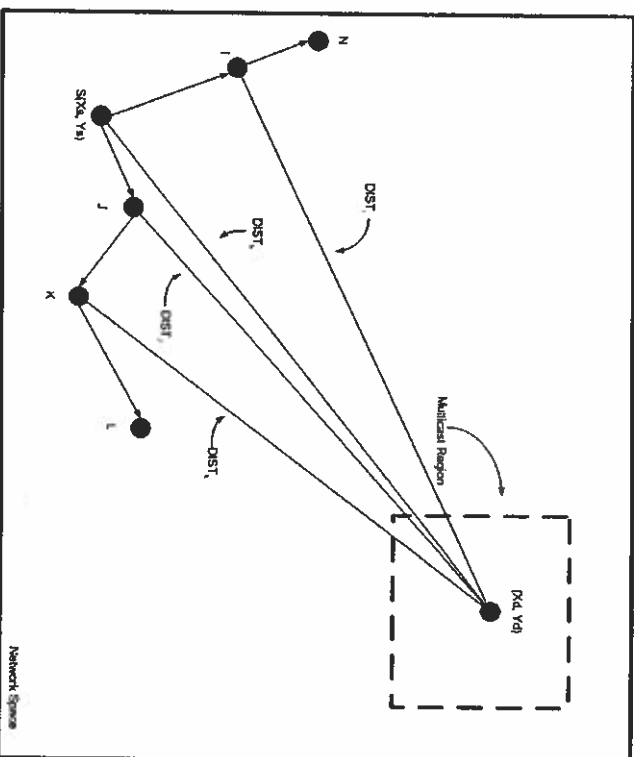


Figure 8: LBM Scheme 2

### 3. Trapezoidal Method

#### 3.1 Approach

My scheme uses a forwarding zone that attempts to have a low overhead, while trying to find a route with a small forwarding zone (low efficiency). We start out by defining the forwarding zone to be trapezoidal in shape, refer to figure 9 to see the first step of the trapezoidal method. I also define the forwarding zone to be small in size, so that it includes the current location of the sender and the multicast region. It should be noted that the sender node  $S$  is capable of determining the coordinates of the forwarding zone. Therefore when node  $S$  transmits a multicast

multicast packet. Consider figure 8, if node  $J$  receives the multicast packet from node  $X$ , all of the steps listed above are performed. Even if node  $J$  is closer to the center of the multicast region than  $X$ , it still forwards the packet. [3]

packet when initiating the multicast delivery, it can include the coordinates of the forwarding zone. Similar to other routing algorithms, when a node receives the multicast packet, it discards the packet if it is not within the region specified by the given coordinates in the packet's header. The size of the forwarding zone depends on the location of the sender as well as the size of the multicast region. To provide control over the size of the forwarding zone, a parameter  $\delta$  is defined, which is used to extend or shift the forwarding zone.

### 3.2 Procedures

As previously stated, my scheme starts out by defining the forwarding zone to be trapezoidal in shape. The first step of the process defines the trapezoid to intersect points A, B, and C of the multicast region. It should be noted that the coordinates of the trapezoid go through the points  $(X_s, Y_s + \delta)$  and  $(X_s + \delta, Y_s)$ . If this shift does not occur then the first step would have a triangle as the forwarding zone. Having a triangle as a forwarding zone would result in a low efficiency. If there happens to be a timeout (the packet can't make it to the multicast region), then the predefined increment  $\delta$  is used to expand the trapezoid. Refer to figure 10 to observe that on the  $n^{\text{th}}$  step A has the coordinates of  $(X_s, Y_s + (n-1)*\delta)$ , and C has the coordinates of  $(X_s + (n-1)*\delta, Y_s)$ .

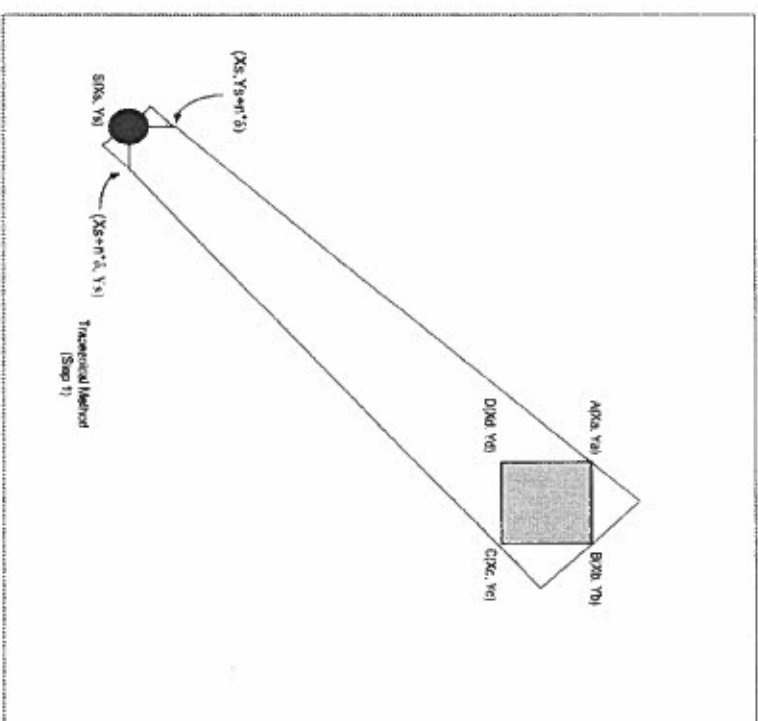


Figure 9: 1<sup>st</sup> step of the Trapezoidal Method





#### 4. Conclusion

This paper focuses on broadcasting to every node within a specified geographical region (geocasting) for mobile ad hoc networking environments. In this paper the multicast region was described as a specified geographical region. Also described was the LBM group, which is the set of nodes that are located within the particular multicast region. In this paper I also discussed two location-based multicast algorithms that limit the forwarding space for a multicast packet to the forwarding zone. Compared to flooding, the algorithms that were discussed resulted in lower message delivery overhead. Also in this paper I proposed a location-based multicast scheme (the Trapezoidal Method) that reduces overhead compared to the previous schemes proposed for location-based multicast environments. For future studies I plan to evaluate the performance of elliptical forwarding zones, as well as other polygon shaped forwarding zones, and compare them to the existing proposed schemes. Another future study is to determine how many timeouts should be taken before other measures are taken.

#### References

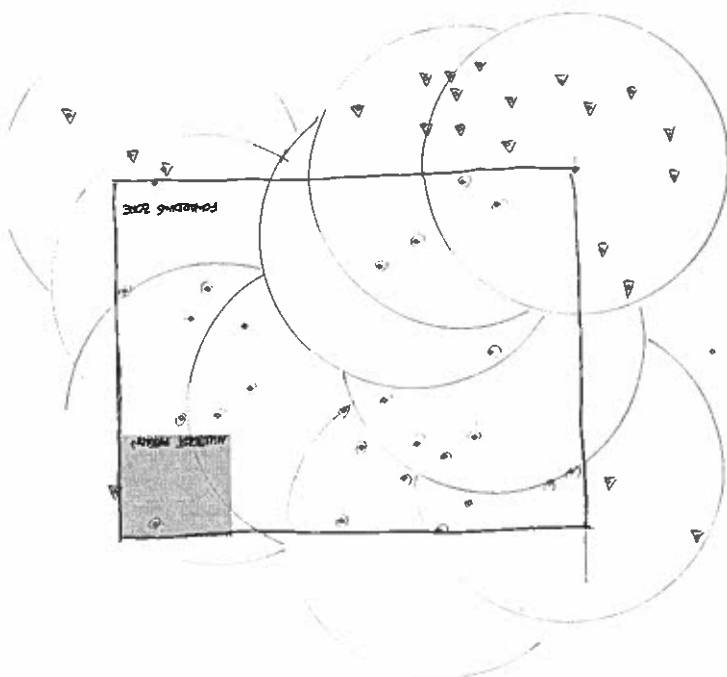
- [1] Y.-B. Ko and N.H. Vaidya. Location-Aided Routing (LAR) in Mobile Ad Hoc Networks. In *ACM/IEEE Int. Conf. on Mobile Computing and Networking (MobiCom '98)*, October 1998.
- [2] Y.-B. Ko and N.H. Vaidya. Location-Based Multicast in Mobile Ad Hoc Networks. Technical Report 98-018, CS Dept., Texas A&M University, September 1998.
- [3] Y.-B. Ko and N.H. Vaidya. Geocasting in Mobile Ad Hoc Networks: Location-Based Multicast Algorithms. *MMCSA 1999*: 101-110.
- [4] C-K Toh. *Ad Hoc Mobile Wireless Networks: Protocols and Systems*. Prentice Hall PTR, 2002.

|     |                                           |                                                |
|-----|-------------------------------------------|------------------------------------------------|
| A-1 | Multicast Flooding                        | Randomly Plotted Nodes 1 (50 nodes)            |
| A-2 | LBM Scheme 1                              | A-3 Trapezoidal Method (1 <sup>st</sup> step)  |
| A-4 | Trapezoidal Method (2 <sup>nd</sup> step) | Randomly Plotted Nodes 2 (50 nodes)            |
| B-1 | Multicast Flooding                        | B-2 LBM Scheme 1                               |
| B-3 | Trapezoidal Method (1 <sup>st</sup> step) | B-4 Trapezoidal Method (2 <sup>nd</sup> step)  |
| B-1 | Multicast Flooding                        | Randomly Plotted Nodes 3 (50 nodes)            |
| C-1 | Multicast Flooding                        | C-2 LBM Scheme 1                               |
| C-3 | Trapezoidal Method (1 <sup>st</sup> step) | Randomly Plotted Nodes 4 (51 nodes)            |
| D-1 | Multicast Flooding                        | D-2 LBM Scheme 1                               |
| D-3 | Trapezoidal Method (1 <sup>st</sup> step) | D-4 Trapezoidal Method (2 <sup>nd</sup> step)  |
| D-5 | Trapezoidal Method (3 <sup>rd</sup> step) | D-6 Trapezoidal Method (4 <sup>th</sup> step)  |
| D-1 | Multicast Flooding                        | Randomly Plotted Nodes 5 (50 nodes)            |
| E-1 | Multicast Flooding                        | E-2 LBM Scheme 1                               |
| E-3 | Trapezoidal Method (1 <sup>st</sup> step) | E-4 Trapezoidal Method (2 <sup>nd</sup> step)  |
| E-5 | Trapezoidal Method (3 <sup>rd</sup> step) | E-6 Trapezoidal Method (4 <sup>th</sup> step)  |
| E-7 | Trapezoidal Method (5 <sup>th</sup> step) | Randomly Plotted Nodes 6 (25 nodes)            |
| F-1 | Multicast Flooding                        | F-2 LBM Scheme 1 (1 <sup>st</sup> step)        |
| F-3 | LBM Scheme 1 (2 <sup>nd</sup> step)       | F-4 Trapezoidal Method (1 <sup>st</sup> step)  |
| F-5 | Trapezoidal Method (3 <sup>rd</sup> step) | F-6 Trapezoidal Method (4 <sup>th</sup> step)  |
| F-7 | Trapezoidal Method (5 <sup>th</sup> step) | Step by Step of the Trapezoidal Method         |
| G-1 | Step 1 of the Trapezoidal Method          | G-2 Step 2 of the Trapezoidal Method           |
| G-3 | Step n of the Trapezoidal Method          | Maple Code                                     |
| H-1 | Maple Code for Random Generated Nodes     | H-2 Com. Maple Code for Random Generated Nodes |

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## LBM SCHEME 1

A-7

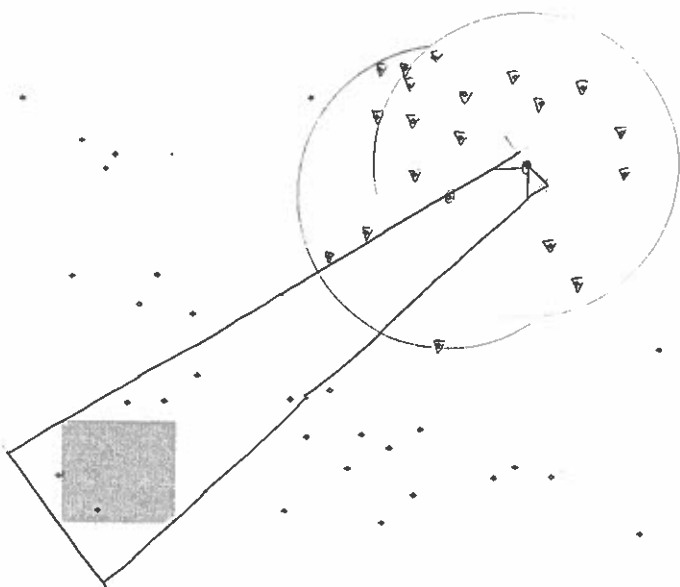


Randomly Plotted Nodes 1 (50 nodes)  
TRANSMISSION COST: 48  
PROCESSING COST: 26

TOTAL COST: 74

## TRAPEZOIDAL METHOD (TM)

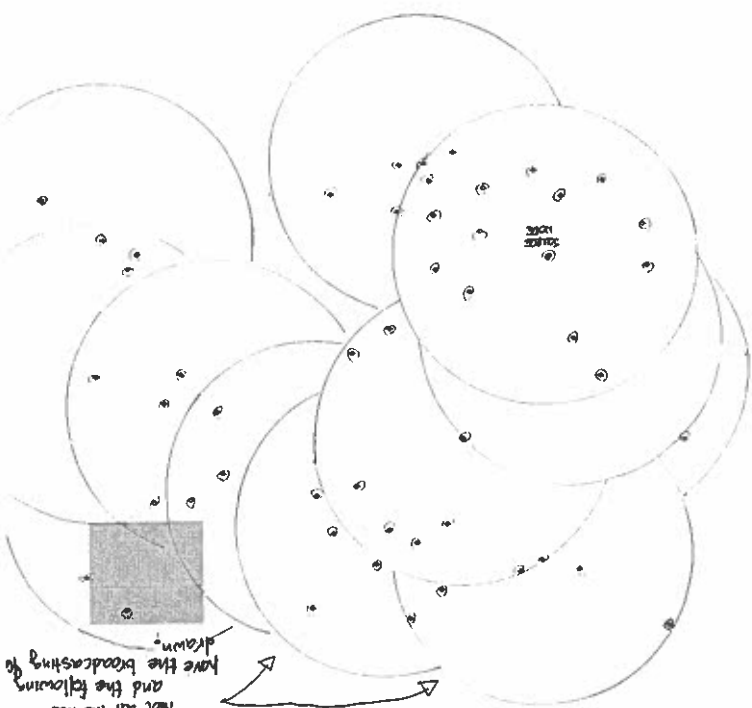
STEP 1



TRANSMISSION COST: 2  
PROCESSING COST: 19

TRANSMISSION COST: 50  
PROCESSING COST: 50  
TOTAL COST: 100

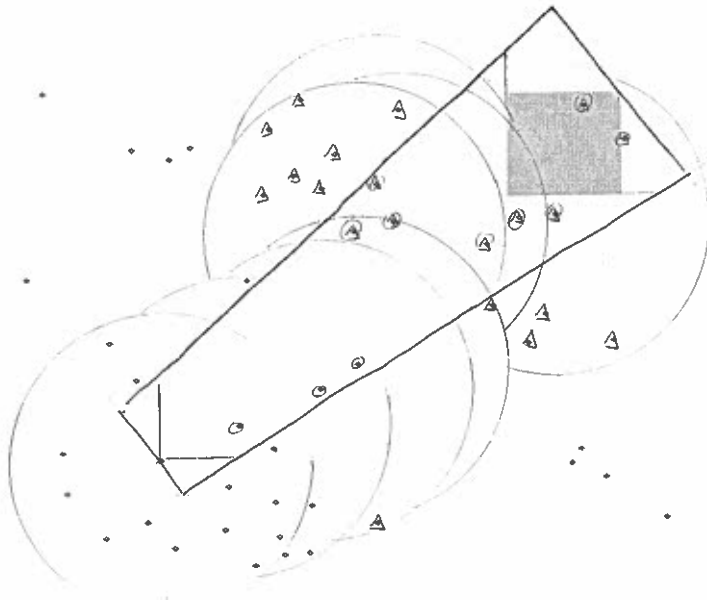
## MULTICAST FLOODING



\* The circles represent the broadcasting range of the nodes on the following page. Not all the nodes have the broadcasting range drawn.

# TM STEP 2

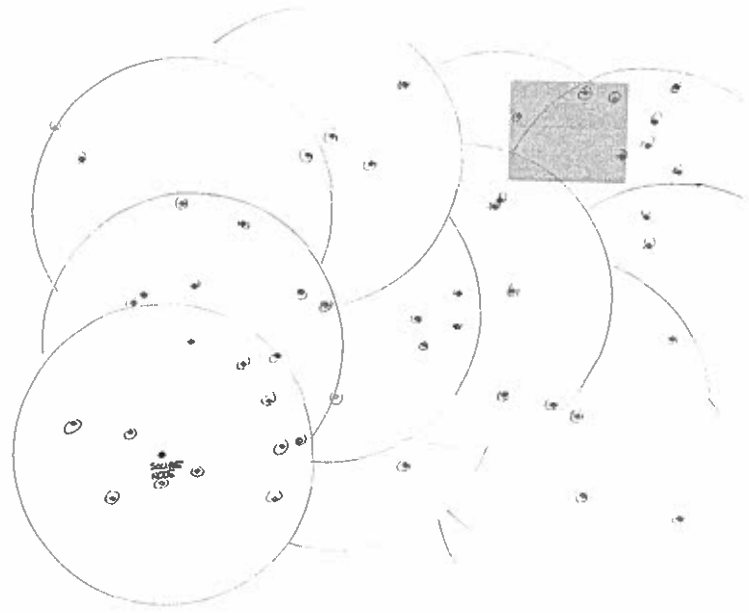
A-4



PROCESSING COST: 20  
TRANSMISSION COST: 11

# MULTICAST FLOODING

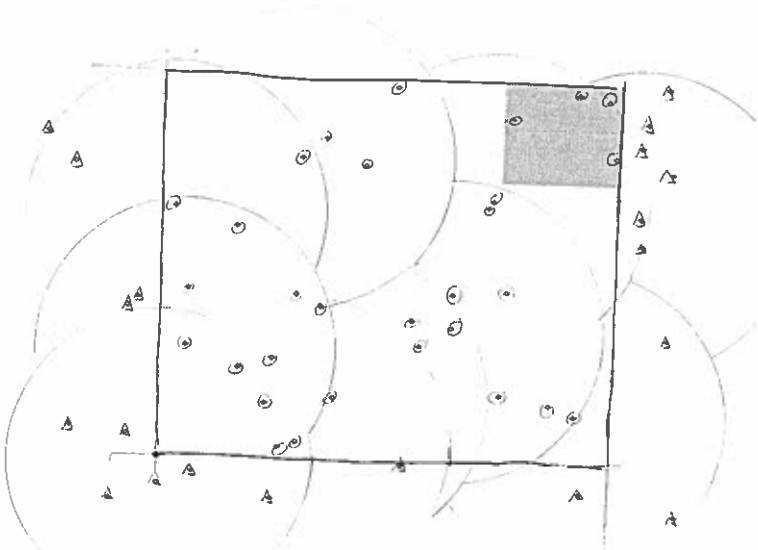
B



FLOODING  
TRANSMISSION COST: 50  
PROCESSING COST: 50  
TOTAL COST: 100

# LBM SCHEME 1

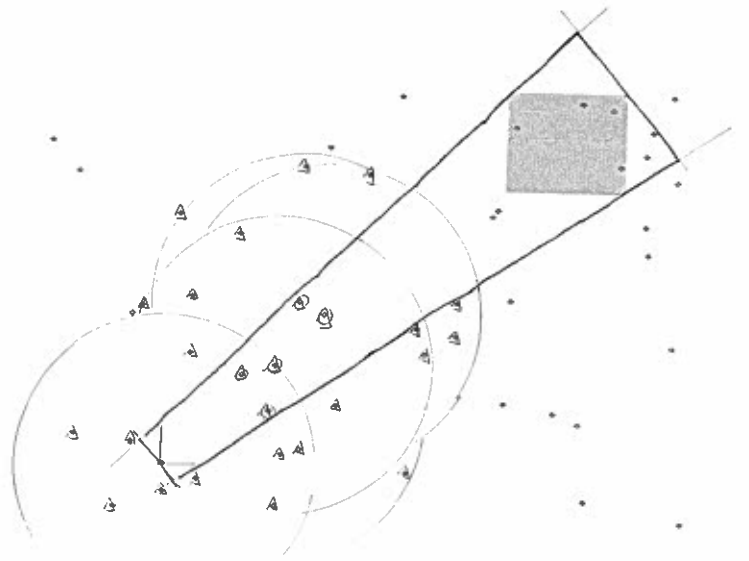
B:



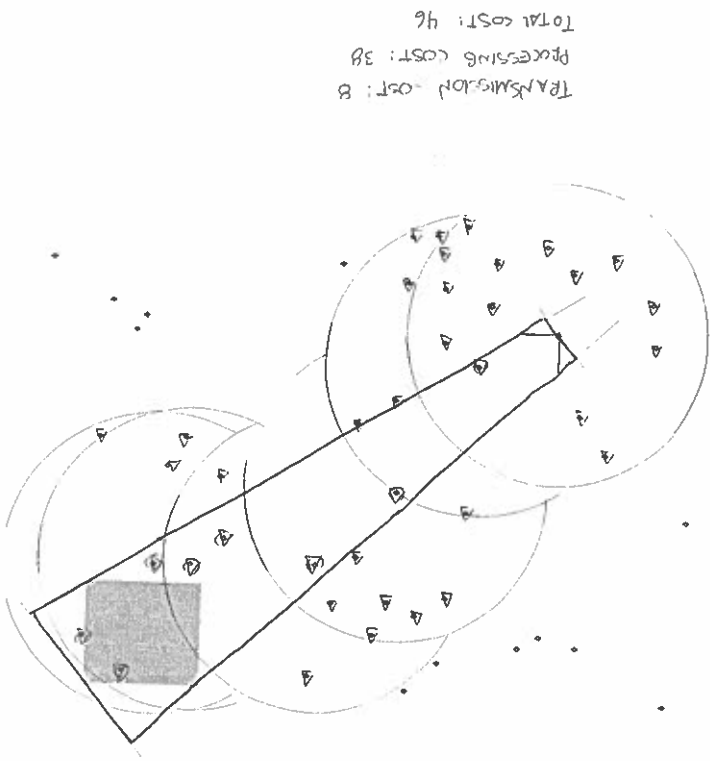
PROCESSING COST: 50  
TRANSMISSION COST: 50  
TOTAL COST: 100

# TM STEP 1

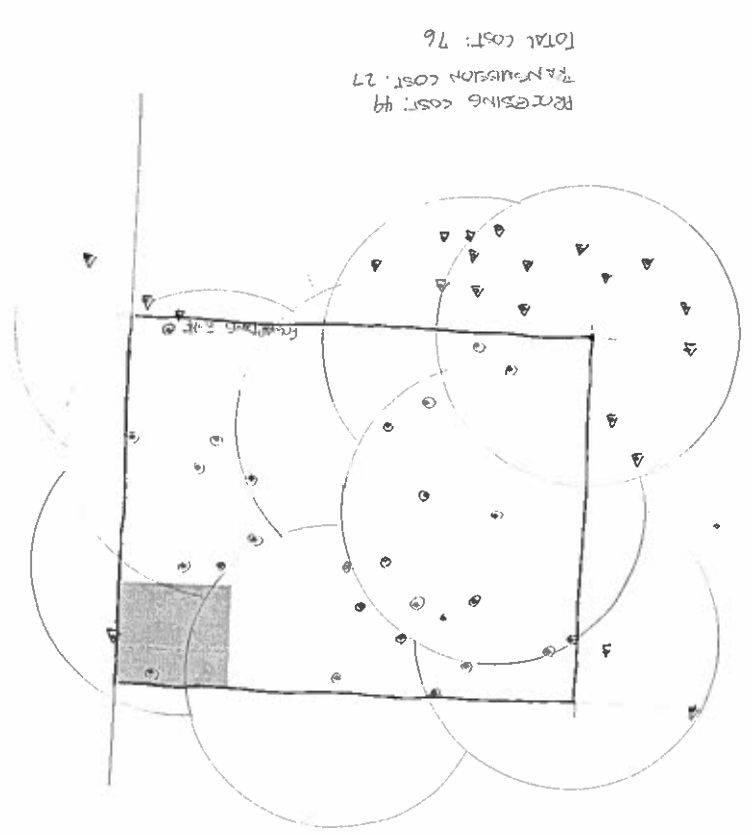
E



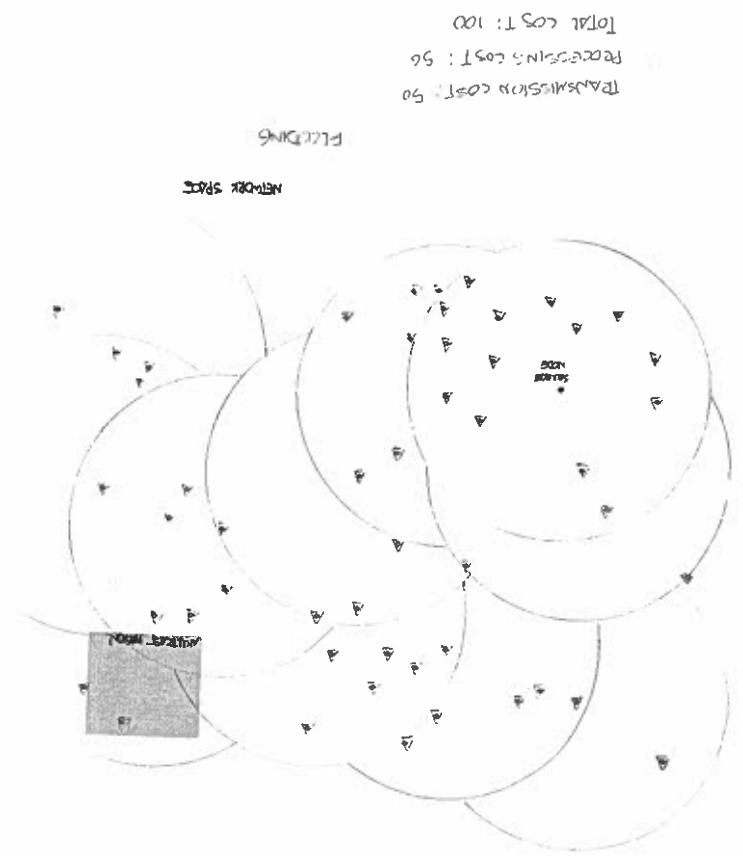
TRANSMISSION COST: 5  
PROCESSING COST: 26



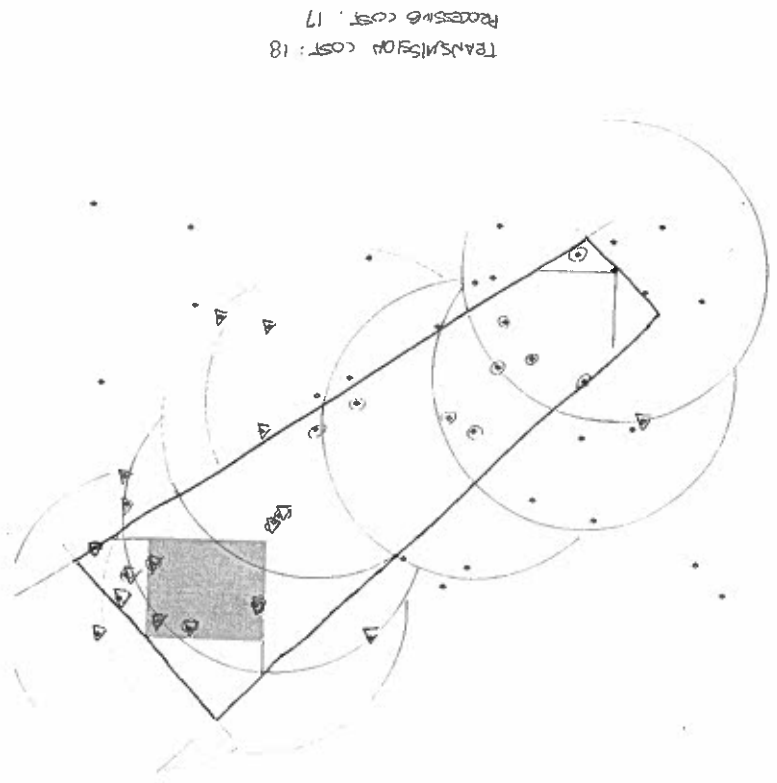
TM STEP 1



LBM SCHEME 1

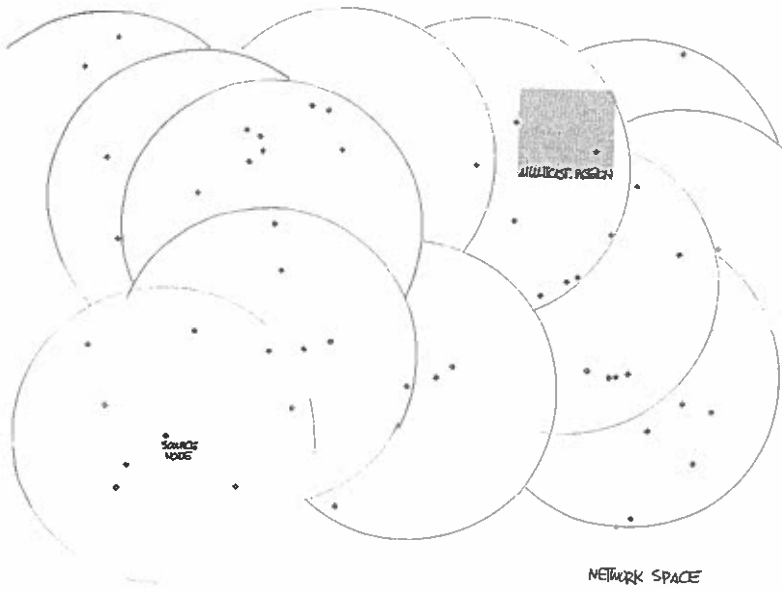


MULTICAST FLOODING



TM STEP 2

## MULTICAST FLOODING

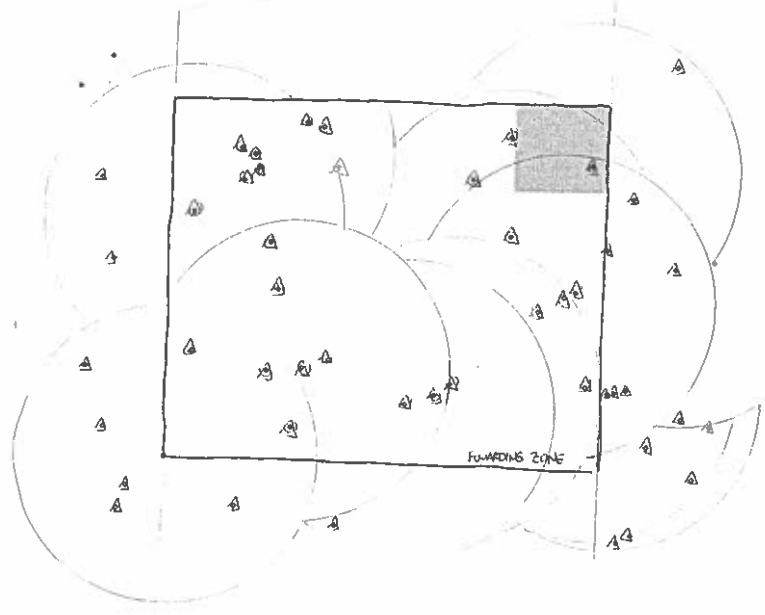


FLOODING:

TRANSMISSION COST: 51  
PROCESSING COST: 51

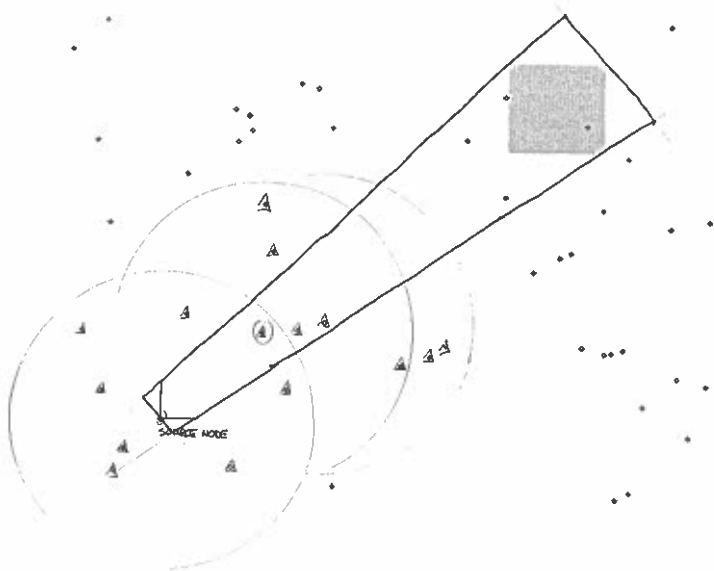
TOTAL COST: 102

## LBM SCHEME 1



PROCESSING COST: 47  
TRANSMISSION COST: 26  
TOTAL COST: 73

## TM STEP 1

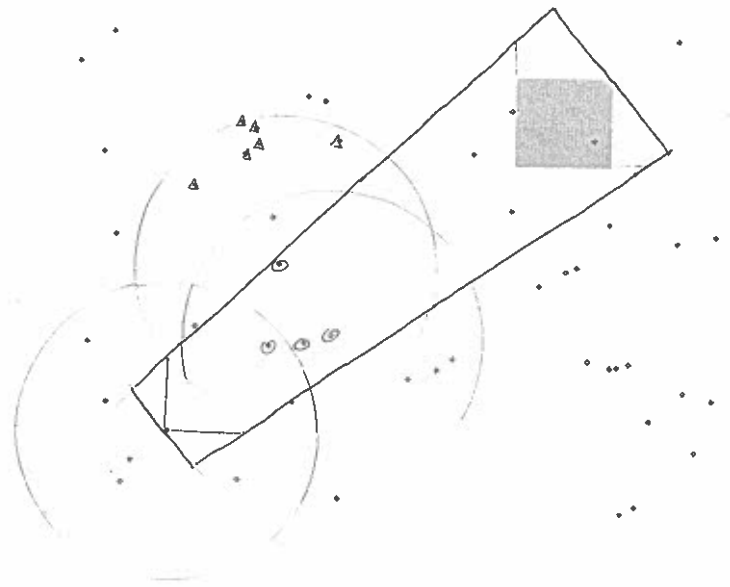


TM METHOD

STEP 1:

TRANSMISSION COST:  $8+5=13$   
PROCESSING COST:  $1+2=3$

## TM STEP 2

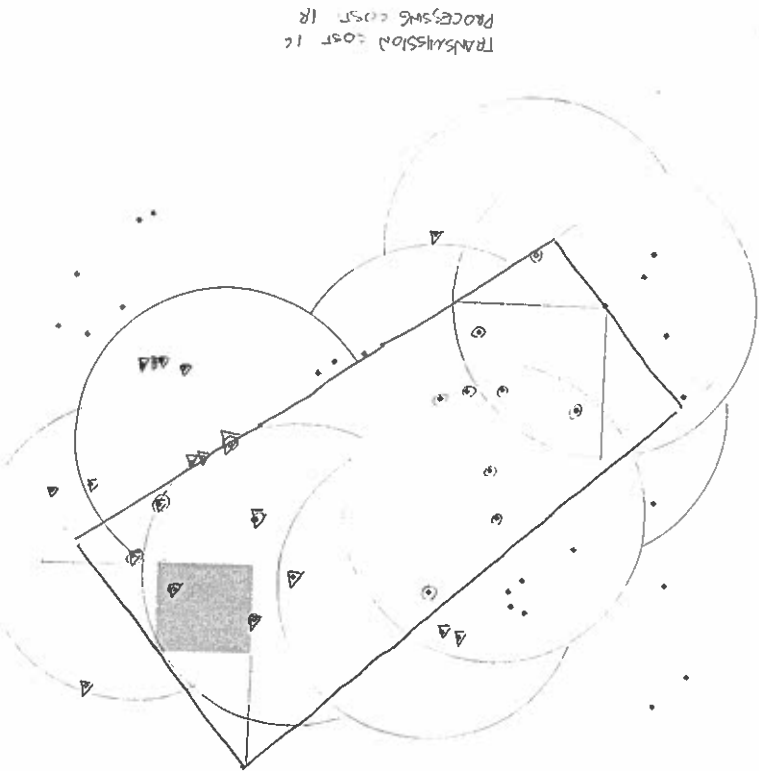


TM METHOD

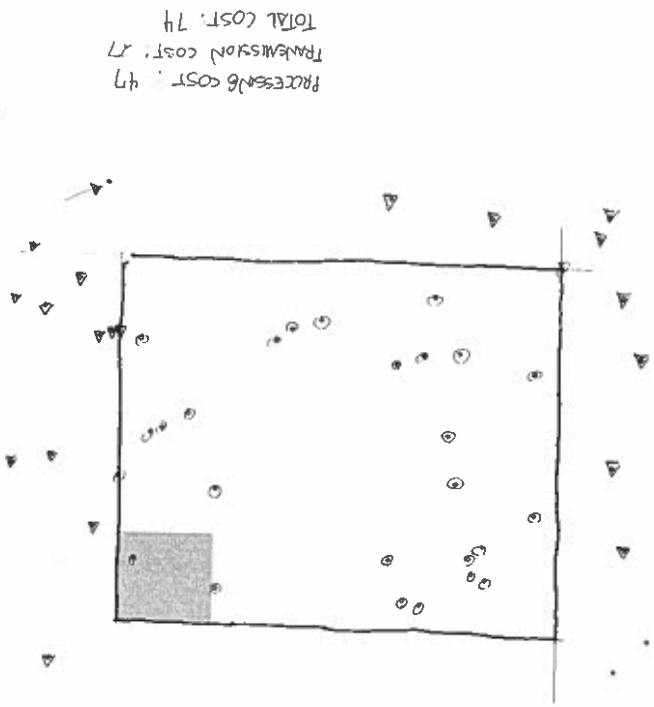
STEP 2:

TRANSMISSION COST: 5  
PROCESSING COST: 6

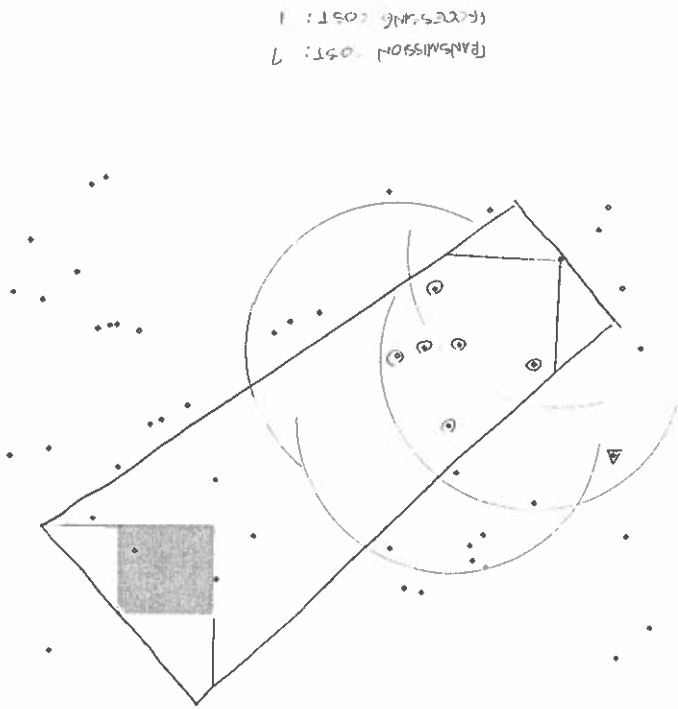
TM STEP 4



LBM SCHEME 1



TM STEP 3



MULTICAST FLOODING

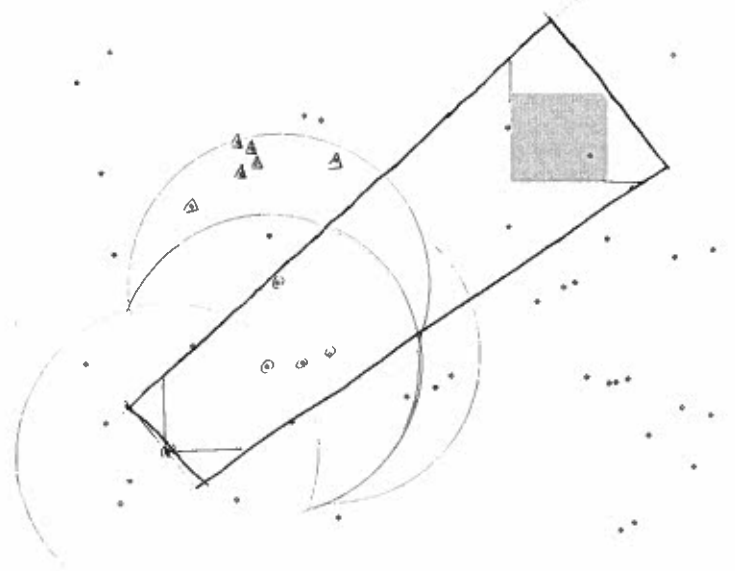
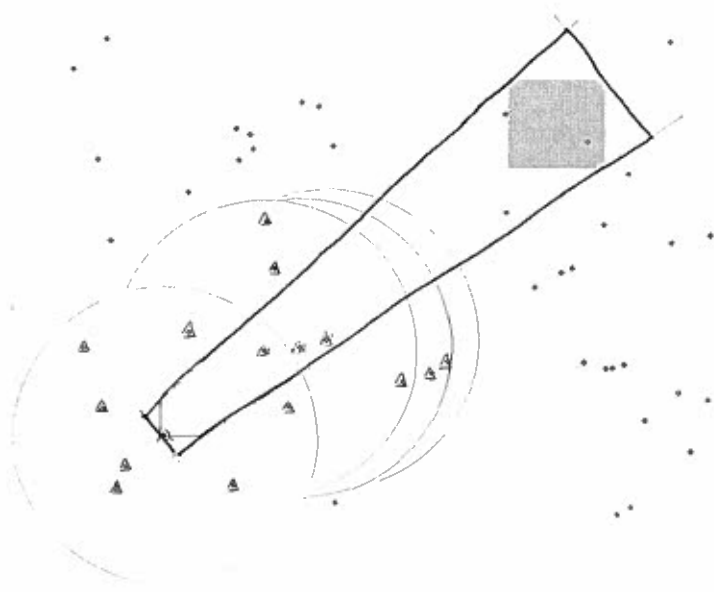


t-3

E-

TM STEP 1

TM STEP 2



TRANSMISSION COST: 4  
PROCESSING COST: 16

TRANSMISSION COST: 5  
PROCESSING COST: 6

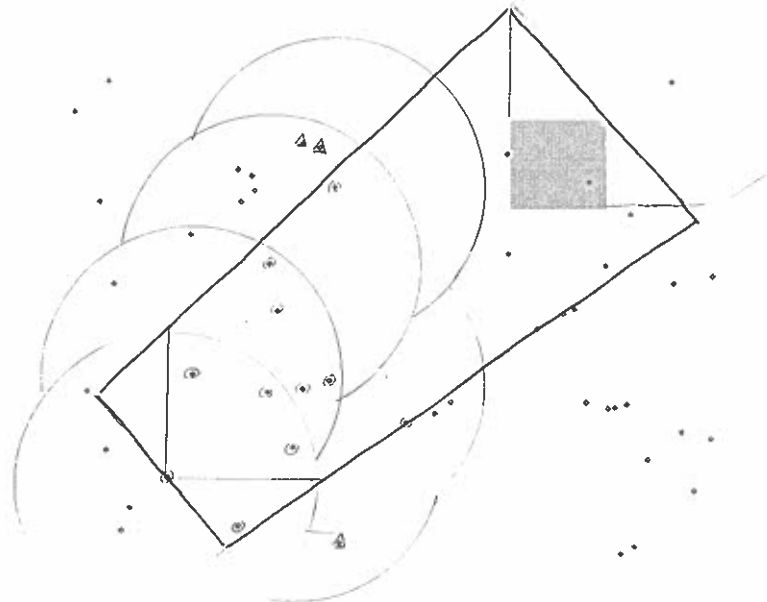
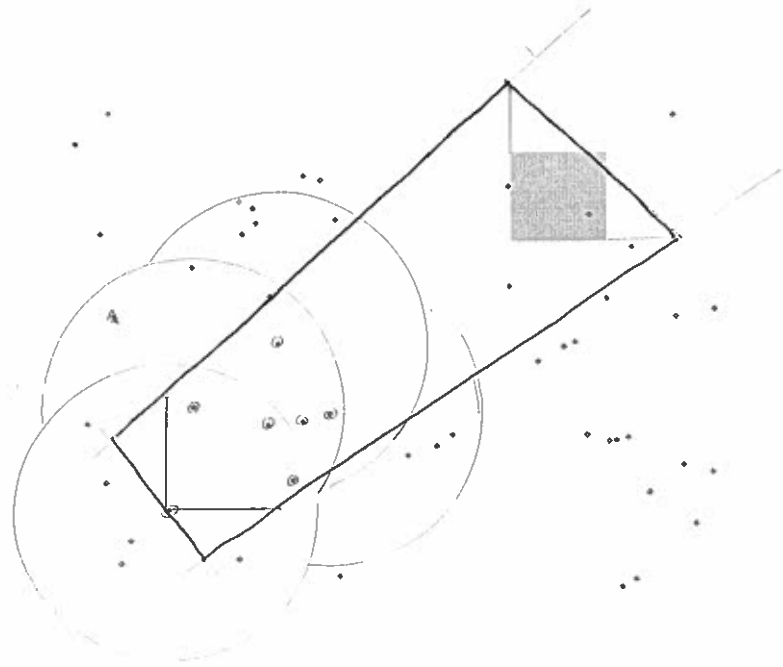
TOTAL COST: 19

E-

E-

TM STEP 3

TM STEP 4

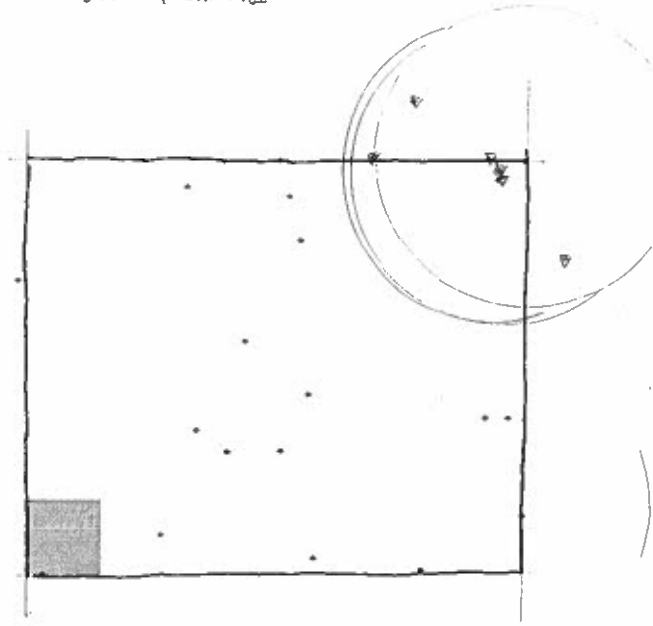


PROCESSING COST: 1  
TRANSMISSION COST: 7

TRANSMISSION COST: 11  
PROCESSING COST: 3



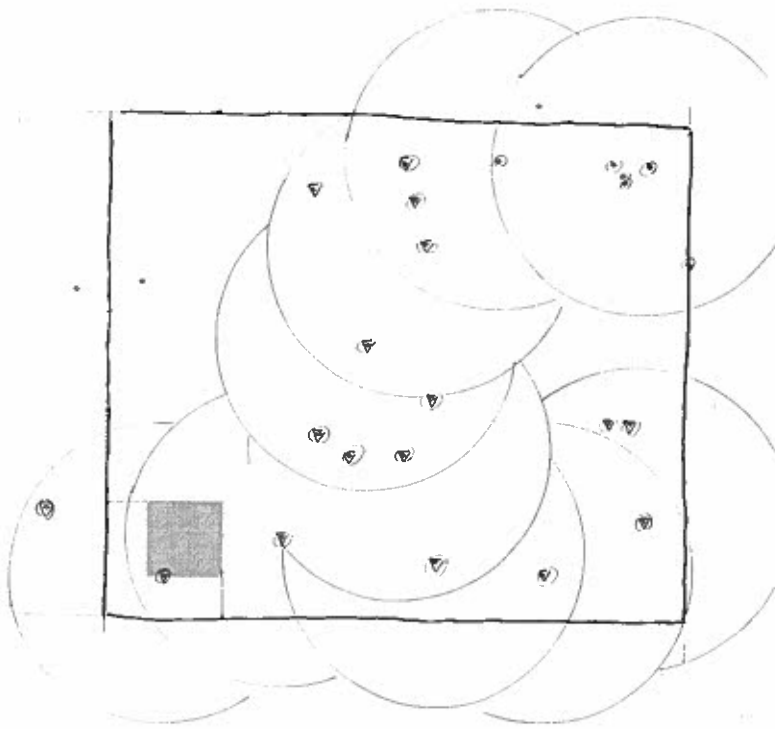
TRANSMISSION COST: 3  
PROCESSING COST: 6



LBM SCHEME 1  
STEP 1

F-2

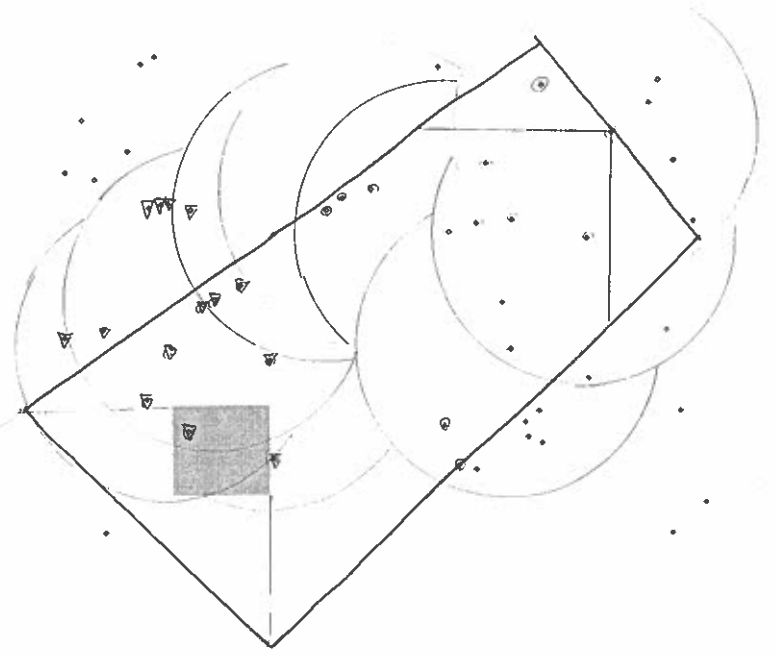
PROCESSING COST: 17  
TRANSMISSION COST: 23



LBM SCHEME 2  
STEP 2

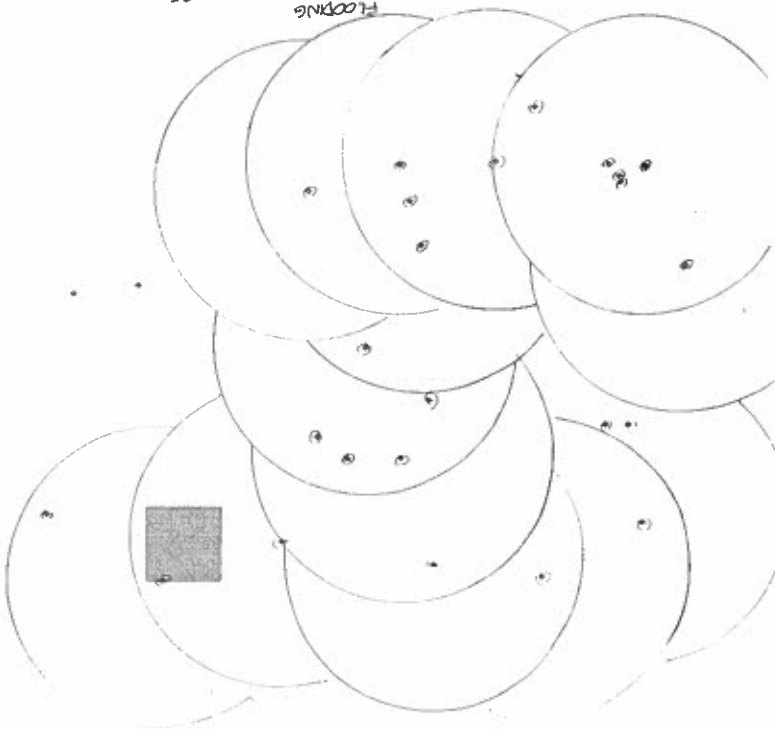
F-1

TRANSMISSION COST: 22  
PROCESSING COST: 14



TM STEP 5

FLOODING  
PROCESSING COST: 25  
TRANSMISSION COST: 25  
TOTAL COST: 50



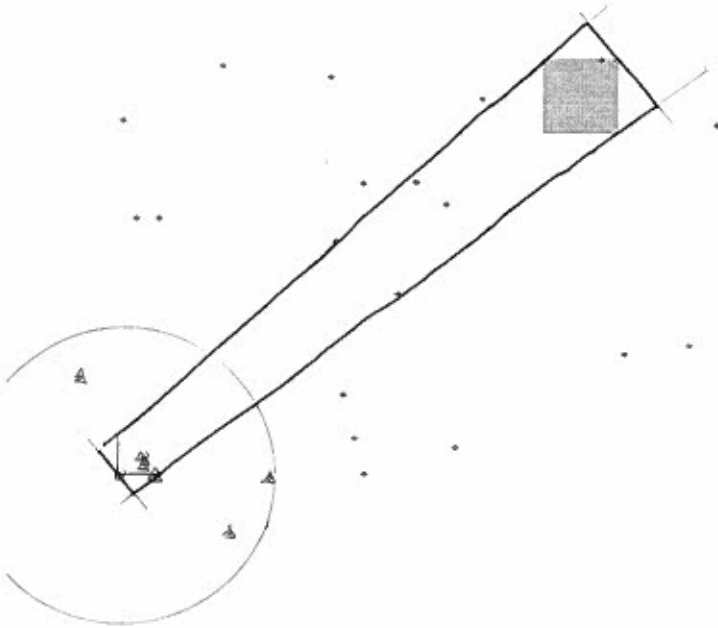
MULTICAST FLOODING

TM STEP 1

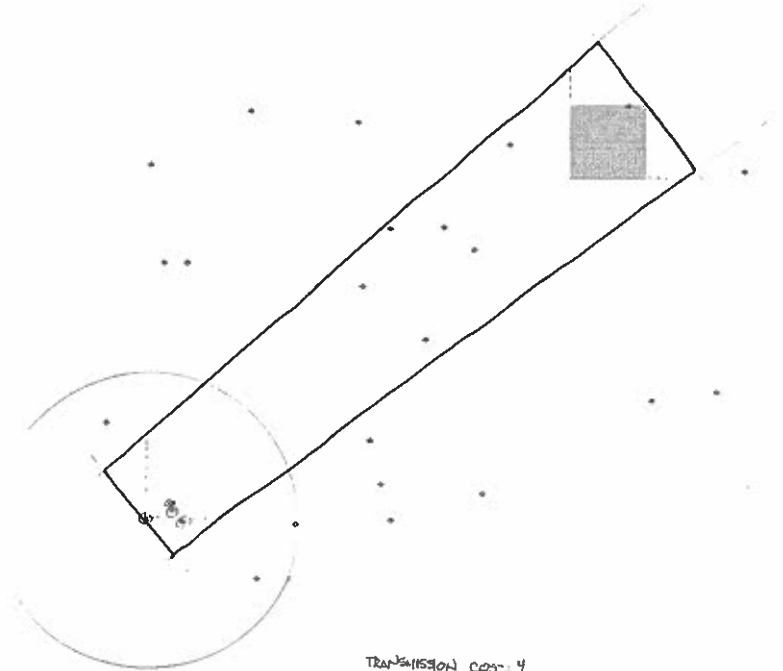
F-4

TM STEP 2

F-



PROCESSING COST: 6  
TRANSMISSION COST: 4



TRANSMISSION COST: 4  
PROCESSING COST: 0

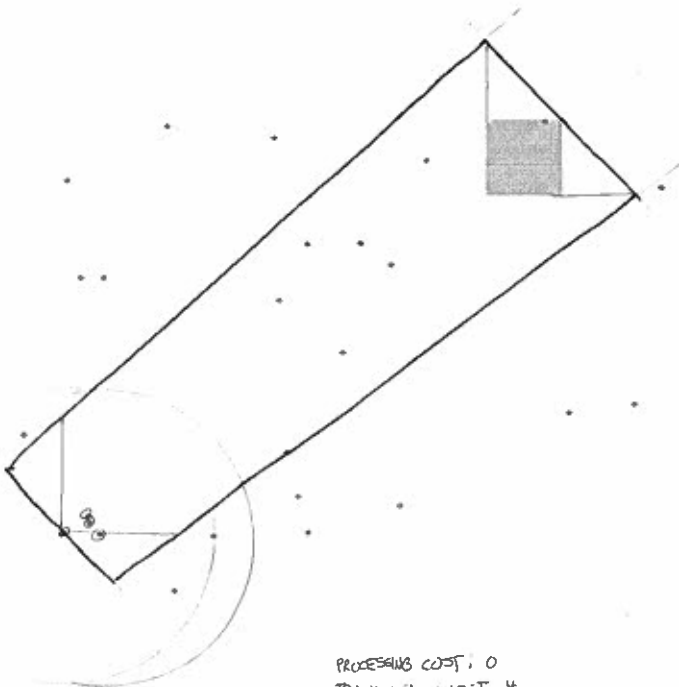
TOTAL COST: 46

TM STEP 3

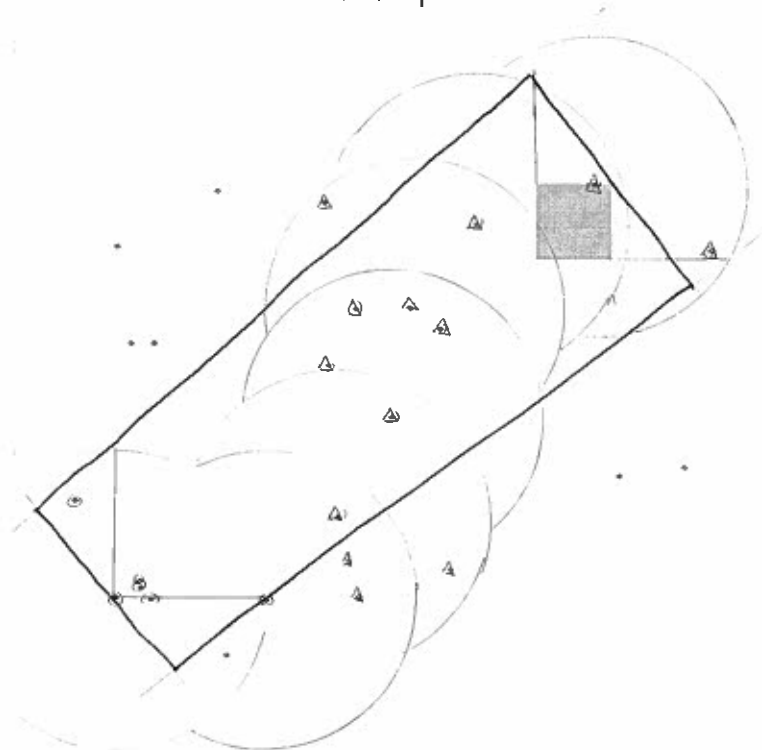
F-6

TM STEP 4

F-7



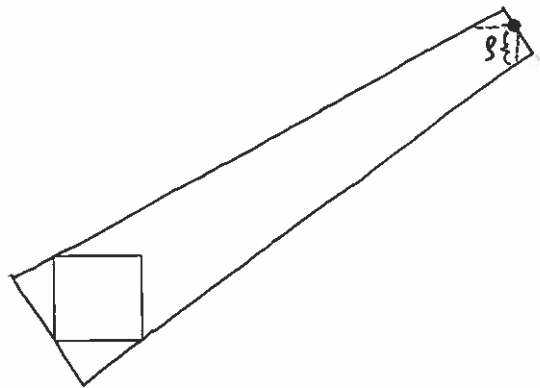
PROCESSING COST: 0  
TRANSMISSION COST: 4



PROCESSING COST: 13  
TRANSMISSION COST: 15

# TRAPEZOIDAL METHOD

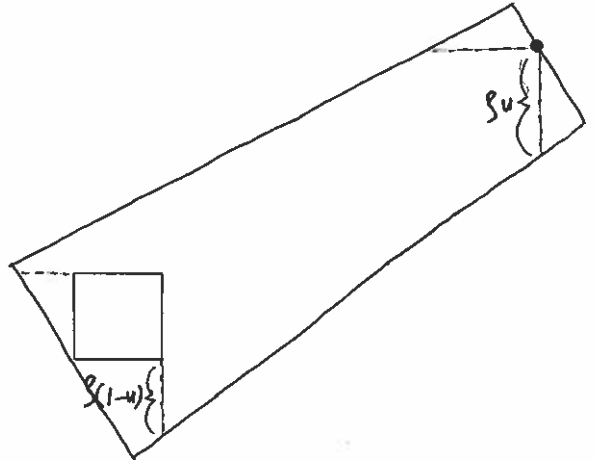
1st STEP



6-3

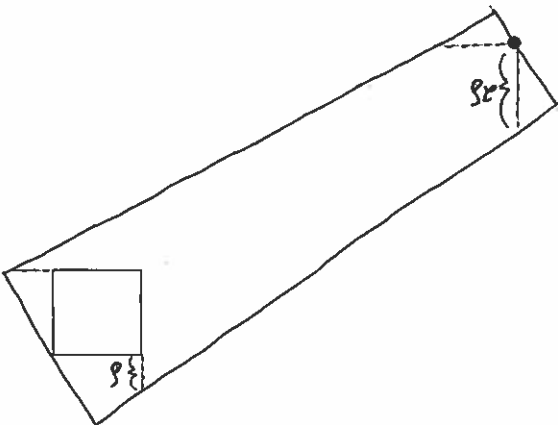
# TRAPEZOIDAL METHOD

nth STEP



2nd STEP

# TRAPEZOIDAL METHOD



MAPLE CODE FOR RANDOMLY GENERATED POINTS

```
> with(stats[scatterplots]):
Warning, these names have been redefined: boxplot, histogram, scatterplot, xscale,
ability, yxexchange, mxexchange, yscale, yheight, yexchange, zscale, zheight
> x_points := [state[random, uniform](1), state[random, uniform](0,
1)] (50);
y_points := [state[random, uniform](1), state[random, uniform](0,
1)] (50);
state[scatterplots, scatterplot](x_points, y_points, color=black);
x_points := [0.8987248808, 0.4855318020, 0.2550506145, 0.9529224743, 0.6420653296,
0.1549126680, 0.85660694384, 0.6814076415, 0.9629177911, 0.8741669464, 0.9059502929,
0.5495528887, 0.08412584224, 0.06706054127, 0.6217577345, 0.2235759057, 0.2735740995,
0.4104243813, 0.8879748579, 0.2344502692, 0.6063862735, 0.07046024115,
0.3220572293, 0.61232201931, 0.6981468984, 0.7885538678, 0.5680466826, 0.9854799213,
0.8105571403, 0.6511981067, 0.2236093056, 0.05971227224, 0.8858054838, 0.2087353384,
0.8634897942, 0.1349121969, 0.8972161318, 0.8253910356, 0.9081688982, 0.8524973108,
0.4252373834, 0.947635354357, 0.3512521904, 0.8716516347, 0.02519936189, 0.1946404552,
0.5485271105, 0.5589958833, 0.3312914882, 0.2568350616]
y_points := [0.6777903170, 0.3263527636, 0.7371619734, 0.5214101451, 0.8357984960,
0.3729093675, 0.3244371666, 0.5327170429, 0.3331280990, 0.772214376, 0.297225555,
0.03513506612, 0.03238723014, 0.04126163242, 0.4648379030, 0.6627421854, 0.052771886036,
0.3642533782, 0.8268041852, 0.4781081141, 0.2221661095, 0.1687044139, 0.6207123021,
0.8528295062, 0.6311640406, 0.6027955568, 0.5642745237, 0.2491072884, 0.054575818658,
0.3292874498, 0.7841174551, 0.4225672762, 0.7454273113, 0.6935178771, 0.6325516909,
0.08707675774, 0.9962856220, 0.1872552414, 0.4689144526, 0.09329956241, 0.8326442705,
0.8767291126, 0.2605821940, 0.06578963842, 0.3169416632, 0.9401000606, 0.5838920174,
0.3423368891, 0.7731227081, 0.6386355489, 0.9050292656]
```

4

CONT. MAPLE CODE FOR RANDOMLY  
GENERATED POINTS

H-2

